

Venkatesh Wadawadagi

+91-9035124315 • venkateshwds@outlook.com • [linkedin.com/in/venkateshwadawadagi](https://www.linkedin.com/in/venkateshwadawadagi)

SUMMARY

Experienced Applied Machine Learning Scientist, technologist and leader with 13+ years of experience, with deep expertise in foundational modeling and large-scale AI/ML systems and leading cutting-edge ML/DL and data science initiatives across diverse domains. Proven ability to coordinate with cross-functional teams, lead and mentor AI/ML, data science professionals, and architect & develop custom algorithms to improve performance and efficiency. Skilled in designing and deploying advanced models/pipelines that push the boundaries of Conversational AI, Generative AI and Visual AI, including large-scale pretraining, and multimodal architectures.

SPECIALIZATIONS

Speech AI • Generative-AI • Computer Vision • NLP • Embedded-AI & IoT • LLMOps •
Statistical Analysis • Data Analytics • Business Intelligence • Video Codecs • Embedded Systems

EXPERIENCE

Sanas

Bangalore, IN

Applied Scientist

Aug 2024 – Present

- Architect, train, and optimize large-scale speech AI models, including speech-to-speech, speech restoration, and speech translation.
- Leveraged self-supervised learning, contrastive learning, and transformer-based architectures (e.g., wav2vec, Whisper, GPT-style models) to improve model accuracy and adaptability.
- Developed efficient model distillation and quantization strategies to deploy large models with low-latency inference.
- Applied active learning, domain adaptation, and synthetic data generation to overcome data limitations.
- Lead efforts in data quality assessment, augmentation, and curation for large-scale training pipelines.
- Designed and implemented scalable model evaluation frameworks, tracking WER, MOS, and latency across diverse scenarios.
- Curated and scaled massive diverse, multilingual, and multimodal datasets for robust model training.
- Work closely with MLOps, Data Engineering, and Product teams to deploy scalable AI systems.
- Ensure seamless integration of foundational models with edge devices, real-time applications, and cloud platforms.
- Translate cutting-edge research into production-grade models that power real-world communication.
- **Tech Stack:** Python, PyTorch, Weights & Biases, Polars, GitLab CI/CD, Hugging Face, ONNX, Docker, FastAPI, AWS, Gradio, Ray, Dagster and PostgreSQL

Rapidflare

Bangalore, IN

AI/ML Consultant

Apr 2024 – Jul 2024

- Mainly worked in the areas of Generative-AI and AI/ML architecture
- Implementation of data ingestion pipeline for product catalog copilot (Question Answering Virtual Assistant)
- Implementation of query routing and query rewriting for product catalog copilot
- Implementation of advanced retrieval techniques to improve performance of RAG for sales copilot

Sabaj AI Software

Bangalore, IN

Principal Data Scientist - Solution Consultant

Nov 2019 – Mar 2024

- Thought leadership and leading practices in the areas of Generative-AI, NLP/LLM, Machine Learning, Deep Learning, Computer Vision & Analytics

- Interfaced with clients, and coordinated with engineering and sales teams to understand & translate business requirements and delivered OKRs by identifying and driving key data science initiatives across multiple industries
- Led and mentored Data Science team to build, deploy and automate end-to-end data science services
- Research and development of open source LLM based QA bot on large corpus of Pdf data. Various RAG evaluation & optimization strategies were experimented and employed to ensure pipeline was reliable and production ready
- Developed a RAG pipeline for text summarisation based Question Answering system for a government entity. Employed metrics such as Faithfulness, Answer Relevancy and ROUGE for reducing hallucinations and improving quality of generated response. Solution was deployed on-premise with optimal latency and memory usage
- Research and exploration in the area of Graph RAG, a retrieval enhancement technique based on knowledge graphs, and demonstrated improved retrieval over baseline RAG
- Developed and deployed a solution for Knowledge Graph creation and Entity Extraction for invoices for one of the leading international bank, where-in custom OCR pipeline was built for text extraction
- Developed Computer Vision based Creative Approvals system for UK's leading media company, thereby ensuring intelligent automation of creative reviews and approval to be compliant with specific guidelines. Our SOTA custom trained models showed an improvement of 18% on Precision and 15% on recall when benchmarked against 3rd party APIs
- Developed a Solution accelerator framework for Deep Learning based Media Analytics. Delivered a solution by leveraging this framework to derive a brand's perceived media value with the help of metrics such as Prominence, Exposure Time etc.
- Developed DNN based Sports Video Analytics pipeline for assessment of athletes in sports schools and achieved 93% accuracy for basketball use case. Ensured models are unbiased and fair, and showcased athlete skill metric generation through Homographic mapping and Semantic analysis
- Mentored various team members in the research and development of challenging and interesting ML/CV problem statements as part of DS Labs, such as Visual Anomaly Detection, Japanese OCR, Video Summarisation and Encrypted Inference
- **Tech Stack:** Python, PyTorch, TensorFlow, Langchain, Haystack, Hugging Face, Pinecone, ChromaDB, Ragas, Docker, FastAPI, DVC, MLFlow, AWS, Gradio, Streamlit, OpenCV, Ffmpeg, NLTK and spaCy

Eagle Eye Networks (formerly Uncanny Vision)

Bangalore, IN

Senior Lead ML Engineer

Jul 2017 – Nov 2019

- Led a team of Data Science professionals and Computer Vision engineers to research, develop, optimize and deploy cutting-edge DL & CV models and systems
- Owning the strategic planning and project management for long term research projects creating value for our customers
- Led a team of data scientists & ML engineers to develop a DL inference framework for low power consumption device, that was aimed to help in the development of next generation e-AI (Embedded-Artificial Intelligence) solutions. Framework was benchmarked against Tegra TX2 to demonstrate improved results for CNNs like ResNet, Tiny Yolo v2 and VGG16 etc. Also, implemented verification environment to do system level and intra-module testing of the framework
- Developed Vehicle Analytics pipeline and employed it to build and deploy solutions on edge & IoT devices by coordinating with SI and Camera vendors. Solutions included Adaptive Traffic Control System and Red Light Violation Detection, and real time testing was done at traffic junctions of Indian cities
- Developed and deployed Worker Safety Monitoring system for one of the leading Indian multinational steel-making companies. Objective was to monitor workers by employing safety gear violation detection & tracking in dangerous environments across industrial premises and ensure safety compliance

- **Tech Stack:** Python, C, Cython, PyTorch, TensorFlow, ONNX, Keras, MXNet, Docker, TensorRT, OpenVINO and Azure DevOPS

Senior Machine Learning Engineer

Dec 2015 – Jun 2017

- Developed various modules of ATM Surveillance system, that was aimed to monitor suspicious activities in ATM premises through activity recognition thereby identifying theft, robbery etc and alerting. Essentially enabling Real-time security for ATMs & customers in remote locations. Solution was deployed on Snapdragon-650 IP camera, that went on to win Design in India Challenge by Qualcomm
- Implemented and integrated the very first alert system on cloud for Uncanny Vision's ATM Monitoring application using AWS cloud services SNS and DynamoDB. Streamlined visualization report generation for Activity Recognition using Microsoft Azure services
- Developed an application to semi-automate the training of DNNs on the cloud based on the images uploaded to AWS S3. Objective was to enable someone without DL knowledge to quickly train, validate & test classifier and object detector models on the desired data set remotely, and also allowing users to download the model once it reaches acceptable accuracy and run it on the mobile phone
- Evaluation of SNPE(Snapdragon Neural Processing Engine) SDK on Snapdragon-650 IP camera. This engagement included activities such as performance analysis, benchmarking, model support and evaluation, mainly targeted to help the vendor with improvements and optimisations
- Developed vision based Smart Refrigerator PoC. Business objective was to identify the person using face recognition running on the tablet attached to a refrigerator which will give JSON output to the embedded target like RZ/G1M where-in bottle detection and categorization were performed to yield the appropriate information(bottle counts) needed from refrigerator for a given person
- Ported UncannyDL(highly optimized DL library from Uncanny Vision) to various embedded devices like Raspberry Pi-2, Renesas RZ/G1E, Mpression Sodra, Snapdragon650 IP-camera and NVidia's Jetson TK1
- **Tech Stack:** Python, C, C++, Caffe, TensorFlow, Keras, OpenCV, Docker, PyQt, SNPE, Boto3, TensorRT, AWS, Azure, Android and PowerBI

Pathpartner Technology

Bangalore, IN

Senior Software Engineer

Jul 2015 – Dec 2015

- Architectural level decision making for software design on ARM subsystem and hardware design on FPGA fabric for HEVC decoder on Zynq-7000
- Project management and mentoring team members across various phases of embedded software development including design, verification, debugging and optimization
- Implementation of pipelined and parallel CABAC for HEVC (H.265) decoder on Zynq-7000
- Built petalinux(Embedded Linux for ARM based FPGAs) v2014.4 and integrated features such as QEMU, JTAG boot etc. Work included various activities such as building linux image, custom driver development, bug fixing and Petalinux datasheet creation using Latex
- **Tech Stack:** C, Verilog, Vivado Toolset, Latex, Linux and Free RTOS

Software Engineer - FPGA Design Engineer

Jul 2012 – Jun 2015

- Streamlined adaptation of Zynq-7000 platform across organization for accelerated computing, by showcasing how SW/HW co-design can help in acceleration through implementation of FIR filter and YUV to RGB conversion on Zynq
- Implemented HEVC(H.265) Decoder on Zynq. Engineered various components including Hardware IP development for Deblocking filter & SAO, Soft IP development for Parser/Scheduler, Implementation of Video DMA for Hardware IPs, programming HDMI & troubleshooting, creating linux image for Zedboard/ZC706, and debugging software/hardware issues

- Implemented cache driver to solve cache issue for data transfer on Zynq-7000, that optimized HEVC decoder performance on Zynq
- Assist team members in understanding Zynq-7000 architecture and resolve design issues in HW and SW portion of Design
- Embedded Linux and Android porting to various embedded targets
- **Tech Stack:** C, Verilog, C++, Linux, Gstreamer, Xilinx ISE, Vivado HLS, Free RTOS

EDUCATION

KLE's B V Bhoomaraddi College Of Engineering & Technology

Hubli, KA, IN

B.Tech in Electronics And Communications Engineering (CGPA: 8.75/10.0)

Aug 2008 - Jun 2012

SKILLS

- History of Strong technical mentorship, managerial and leadership experience, ownership and accountability
- Take initiatives to launch projects with the ability to quickly learn and develop own ideas in an interdisciplinary environment
- A strong record of prior research, product development and consulting experience
- A Research-to-Implementation Mindset: ability to spearhead a project from idea to experimentation to prototype to implementation
- Strong quantitative analysis, programming, and statistical modeling skills
- Interest and ability to quickly learn & apply newer machine learning algorithms
- Excellent communication and presentation skills. Proven ability to interact with all levels of the organization including senior leadership and executives
- Strong ability to manage multiple projects for multiple stakeholders and manage expectations
- Strong organizational skills and attention to detail
- Good problem solving and analytics skills
- Strong communication skills and the ability to explain complex solutions to a diverse non-technical audience and business users
- Knowledge sharing internally and externally through conferences, journals, blogs and articles

CERTIFICATIONS

- 3D Vision Summer School | IIIT Hyderabad
- Marketing Analytics | University of Virginia Darden School of Business
- AWS Machine Learning Foundation Course | Udacity

PUBLICATIONS

- Efficient and Optimal Deep Learning Inference for Computer Vision Applications | Lattice - The Machine Learning Journal by AdaSci | Volume 2, Issue 1 | Feb 22, 2021
- Metrics to Use to Evaluate Deep Learning Object Detectors | KD Nuggets | Aug, 2020
- Enabling VDMA-based fast data transfers on ZYNQ | Aug, 2015
- Design AXI Master IP using Vivado HLS tool | Sep, 2014

SPEAKING ENGAGEMENTS

- The Promise and Potential of AI/LLM in Decentralized Finance | DevDay | Feb-2024 | Chennai, IN | <https://devday.sahaj.ai/chennai/defi/home>
- Shipping LLM: Addressing Production Challenges | Pune's 50th DevDay | Jan-2024 | Pune, IN | <https://youtu.be/HxRDyMB87dI?si=Zklysixbch8Jagw>

- Uncovering Business Potential with Deep Learning Based Media Analytics | AI Summit London | Jun-2023 | London, UK | <https://1drv.ms/v/s!Aq7WEsxcwBdp8WJEXj-HG0fkZynX?e=OULi5o>
- Vision based Brand Analytics | DPhi's (AI Planet) Saturday Live Session | Aug-2022 | <https://www.youtube.com/live/1OCSvNXfaFM?si=GTeqAQ2LPSEnc41D>
- AI based Worker Safety Monitoring for an Industrial Site | AI Expo by ValleyML | Oct-2021 | <https://vimeo.com/639240028/8a4a5b9894>
- Challenges in Vision based Adaptive Traffic Control Systems | Embedded Vision Summit | May-2021 | <https://youtu.be/0NBDrlSKPAM?si=FL6ttB1Br7NIlvkv>
- Use Case - Accelerating AI on Surveillance Camera for Worker Safety Monitoring | Software for AI Optimization Summit | Jun-2021 | <https://vimeo.com/556098249/70d43c23d6>
- Efficient and Optimal Deep Learning Inference for Computer Vision Applications | MLDS | Feb-2021 | <https://mlds.analyticsindiamag.com/speaker/venkatesh-wadawadagi/>

BLOGS / ARTICLES

- TensorFlow vs PyTorch: Deep Learning Frameworks | Knowledge Hut | <https://www.knowledgehut.com/blog/data-science/pytorch-vs-tensorflow>
- Accelerating Inference: Neural Network Pruning Explained | Paperspace | <https://blog.paperspace.com/neural-network-pruning-explained/>
- Definitive Guide on Data Science Modeling | Knowledge Hut | <https://www.knowledgehut.com/blog/data-science/data-science-models>
- AI-based Vehicle Analytics for Smart Cities | Medium | <https://medium.com/inspiredbrilliance/ai-based-vehicle-analytics-for-smart-cities-2c795681d077>
- Data Set and Data Augmentation for Face Detection and Recognition | Hackernoon | <https://hackernoon.com/data-set-and-data-augmentation-for-face-detection-and-recognition-6eu3n0a>
- AI for Manufacturing Industry | AI Time Journal | <https://www.aitimejournal.com/ai-for-manufacturing-industry/19015/>
- 7 Real-World Applications of AI in Healthcare | Hackernoon | <https://hackernoon.com/7-real-world-applications-of-ai-in-healthcare-eu15031jt>